

Class 36: The Gram-Schmidt Process

Quick Review: An orthogonal matrix, Q , is

Two nice geometric properties of any orthogonal matrix Q :

For any vectors $\mathbf{x}$ and $\mathbf{y}$, $Q\mathbf{x} \cdot Q\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$

1. This means that multiplying a vector by Q will not change the length of the vector.
2. This also means that an orthogonal transformation will maintain the angle between any two nonzero vectors.

Big Question: How do we turn an arbitrary set of linearly independent vectors into an orthonormal set with the same span?

Practice Problem: Find an orthonormal basis for the column space of the matrix

$$A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix} = [\vec{a}_1 \quad \vec{a}_2 \quad \vec{a}_3]$$

$$\text{Let } W = \text{col}(A) = \text{span}\{a_1, a_2, a_3\}$$

First find orthogonal vectors $\{v_1, v_2, v_3\}$

such that $\text{span}\{v_1, v_2, v_3\} = \text{span}\{a_1, a_2, a_3\} = W$

Finally divide each v_i by its length to create an orthonormal basis. $Q = \begin{bmatrix} \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \end{bmatrix}$

QR Factorization: In many applications it is useful to write an $m \times n$ matrix, A , having linearly independent columns as $A = QR$ where Q is an $m \times n$ matrix with orthonormal columns and R is $n \times n$, upper triangular, and invertible.

Three step process to find the QR factorization for any matrix A with linearly independent columns.

Step 1.

Step 2.

Step 3.

Practice: Find the QR factorization for the matrix in the previous example.

$$A = \begin{bmatrix} 1 & 3 & 5 \\ -1 & -3 & 1 \\ 0 & 2 & 3 \\ 1 & 5 & 2 \\ 1 & 5 & 8 \end{bmatrix} \quad Q = \begin{bmatrix} \frac{1}{2} & \frac{1}{2\sqrt{2}} & \frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2\sqrt{2}} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{2} & \frac{3}{2\sqrt{2}} & \frac{1}{2} \\ \frac{1}{2} & \frac{5}{2\sqrt{2}} & \frac{1}{2} \end{bmatrix} \quad Q^T = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{3}{2\sqrt{2}} & \frac{5}{2\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

Verify that $QR = A$